

KKK OIL FACTORY

Enterprise ERP & Industrial Billing System

Comprehensive Architecture, Manufacturing Operations & User Manual

Document Overview & Technical Release

This technical document serves as the authoritative operational and development reference for the KKK Oil Factory ERP system. The platform unites raw oilseed procurement, expeller crushing yield tracking, dual GST / Non-GST invoicing, inventory control, and logistics management across Web, Android, and iOS devices.

Key System Highlights

- Production Crushing: Real-time calculation of seed input vs oil yield %, cake %, and loss %.
- Statutory Billing: Rule 46 CGST compliant invoicing with HSN 1515/2306 tax splits.
- Fast Counter POS: Rapid retail counter billing with cash change and 80mm thermal printing.
- Offline Synchronization: Local disk caching with Google Cloud Firestore sync.
- Cross-Platform: Single reactive Dart codebase powering Web, Android, and iOS.

SYSTEM METADATA:

- Enterprise: KKK Oil Factory, Tamil Nadu, India
- Version: 2.4.0 (Production Release)
- Target Framework: Flutter 3.x / Dart 3.12+
- Database Engine: Cloud Firestore + SharedPreferences
- Document Classification: Proprietary Enterprise Documentation

1. Technical Architecture & Security

The application utilizes a decoupled, reactive architecture built on Flutter Provider. State changes from field operations immediately propagate through data providers to connected UI widgets.

State & Storage Pipeline

- AuthProvider: Manages cryptographically hashed user credentials, JWT session tokens, and module permission trees.
- DataProvider: Maintains active factory state including product lists, silo dip levels, batch yields, customer balances, and pending dispatch routes.
- StorageService: Preserves user sessions across browser reloads and URL changes via encrypted key-value pairs.
- HTML5 Clean Routing: Direct URL routing (/dashboard, /billing, /inventory) with built-in route guarding.

Role-Based Access Control (RBAC)

- Super Admin: Unrestricted access (*) across master settings, financial audits, GST/Non-GST toggle, and employee management.
- Admin: Manages product master catalogs, retail/wholesale pricing rules, and factory operations.
- Manager: Oversees cashier shifts, validates sales orders, and monitors batch yields.
- Cashier: High-speed retail counter billing, B2B invoicing, and receipt printing.
- Inventory Incharge: Tracks silo tank dips, packaged SKU counts, and packaging materials.
- Production Supervisor: Logs seed intake, expeller running hours, and oil/cake yields.
- Driver: Manages trip sheets, deliveries, and digital customer acknowledgments.

2. Billing Engines & Expeller Operations

The ERP incorporates two distinct commercial engines optimized for retail throughput and formal wholesale tax compliance.

Dual Billing Workflow

- Counter POS: Single-tap checkout with auto change calculator and ESC/POS thermal printing.
- B2B Tax Invoicing: Complete GST calculations based on customer state code:
 - Intra-State (Tamil Nadu 33): CGST 2.5% + SGST 2.5% = 5.0% total tax.
 - Inter-State: IGST 5.0% flat applied to tax-registered out-of-state buyers.
- HSN Code Mapping: HSN 1515 for finished edible oils, HSN 2306 for oil cake byproducts.
- Payment Flexibility: Cash, UPI QR code generation, Bank NEFT/RTGS, and Ledger Credit.

Expeller Extraction Formulas

MASS BALANCE RECOVERY EQUATIONS:

Oil Yield % = (Filtered Oil Output [kg] / Seed Input [kg]) * 100

Cake Yield % = (Oil Cake Produced [kg] / Seed Input [kg]) * 100

Processing Loss % = 100% - (Oil Yield % + Cake Yield %)

Groundnut Baseline: 1,000 kg Seeds -> 400-420 kg Oil (41%) + 540-560 kg Cake (55%)

3. Logistics, Compliance & Deployment

Comprehensive supply chain tracking connects weighbridge seed arrivals to customer deliveries.

Procurement & Logistics Modules

- Mandi Intake: Records gross, tare, and net weights with moisture deduction percentages.
- Supplier Ledger: Complete vendor accounts payable and payment disbursement schedules.
- Dispatch Management: Route scheduling, vehicle registration, and driver trip notes.
- E-Way Bill Support: Records 12-digit E-Way Bill numbers for consignments exceeding Rs. 50,000.

Production Build & Deployment Commands

Commands to compile production-ready release artifacts across supported platforms:

- Web (CanvasKit): `flutter build web --release --web-renderer canvaskit`
- Firebase Hosting: `firebase deploy --only hosting`
- Android APK: `flutter build apk --release`
- Android App Bundle: `flutter build appbundle --release`
- iOS Xcode Build: `cd ios && pod install && cd .. && flutter build ios --release`

Standard Shift Closing Procedure

- 1. Verify all counter sales orders are closed and marked as Paid.
- 2. Navigate to Reports -> Daybook to view shift cash and digital totals.
- 3. Count physical cash in drawer and reconcile with system cash balance.
- 4. Super Admin or Manager signs off on shift closing report.